

Sanjay Kumar Samala

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EDUCATION

Master of Science in Digital Forensic George Mason University, Fairfax, VA Relevant Coursework: Networks & Cybersecurity, Pen Testing, Malware Reverse Engineering, Cloud Security, Mobile Forensics	Jan 2024 – Dec 2025 CGPA: 3.44
Bachelor of Engineering in Electrical and Electronics Osmania University, Hyderabad, India	Aug 2019 – Jul 2023 CGPA: 3.55

SKILLS

Offensive Security:	Nmap, Hydra, Burp Suite, Metasploit , Nessus, Vulnerability Assessment, Web Exploitation
Defensive Security:	SIEM, Splunk , Zeek, Wireshark , Incident Response, Threat Modeling, OWASP Top 10
Forensics & Malware:	FTK Imager, Autopsy , IDA Pro, Hex Editor , Process Monitor, Process Explorer, Forensic Chain of Custody
Programming:	Python, C

EXPERIENCE

Research Assistant - Mason Living Innovation Lab George Mason University	Summer 2025 – Present Fairfax, VA
- Conducting embedded hardware security research focused on Cyber-Physical Systems (CPS) and industrial automation environments. - Hands-on experiments using ChipSHOUTER EMFI tool and Rigol oscilloscope to study device behavior under fault injection. - Collecting and analyzing timing and power signals from embedded devices during controlled tests. - Setting up lab equipment, calibrating tools, and writing Python scripts to record and organize results.	
Graduate Teaching Assistant George Mason University	Aug 2024 – Present Fairfax, VA
- Leveraging the OSI reference model and advanced data link protocols, orchestrated interactive demonstrations of packet switching and congestion control. - Utilizing Burp Suite, Kali Linux, and Metasploit, directed offensive security simulations and facilitated exploit development labs, enhancing technical competence. - Applying SSL/TLS configuration principles with Zeek log analysis and Splunk, engineered secure communication environments for 5+ web applications, reducing exploitable vulnerabilities in encryption practices.	
Penetration Testing Intern HackersForYou	Sep 2023 – Oct 2023 Hyderabad, India
- Performed penetration testing and web application security assessments using Nmap, Burp Suite, and Metasploit to identify vulnerabilities such as privilege escalation and buffer overflow. - Automated reconnaissance and vulnerability validation with Python scripts, reducing manual testing time by 25% and accelerating remediation workflows. - Documented findings with CVSS scoring and aligned remediation strategies with OWASP Top 10 and MITRE ATT&CK frameworks to strengthen system security and incident response.	

Research Intern – Research Centre Imarat (RCI), DRDO

- Conducted simulation research on **Brushless DC motor** control systems using **MATLAB-Simulink**, integrating concepts of **cyber-physical systems** security, threat modeling for control logic, and anomaly detection in motor telemetry, which enhanced resilience strategies for critical infrastructure and defense-grade embedded systems.
- Developed fault detection and control optimization models in **MATLAB/Simulink** for BLDC motors, using signal processing and system identification techniques to analyze dynamic behavior and enhance system reliability.

ACADEMIC PROJECTS

Malware Reverse Engineering

- Using static analysis with Hex Editor, **IDA Pro**, and **YARA** rules, examined PE headers, disassembled binaries, and IOC strings, which uncovered **15+ critical vulnerabilities**, stealth persistence mechanisms, and malicious API calls.
- Employing dynamic analysis via Process Monitor, Process Explorer, and **Wireshark**, tracked system behavior to reveal **12 unauthorized registry modifications**, suspicious DLL injection attempts, and anomalous file operations, strengthening forensic understanding of the malware’s evasion tactics.
- Applying reverse engineering workflows with **FakeDNS** and execution flow mapping, intercepted malicious C2 DNS requests, correlated behavioral anomalies, and developed targeted countermeasures, achieving a **60% reduction** in malware infection risk through proactive remediation strategies.

Practical Hard-Drive Forensic Examination – Expert Witness Report

- Using **FTK Imager** with MD5/SHA-1 verification and WiebeTech UltraDock write-blocker, acquired a forensically sound image of a 40 GB Seagate Barracuda HDD, resulting in validated bit-level accuracy and preservation of chain of custody.
- Leveraging **Autopsy** 4.22.1 with metadata, file carving, and browser artifact analysis, recovered emails, financial spreadsheets, and jailbreak traces, which tied suspects to iPhone resale operations and revealed **19 flagged CSAM images** and **14 adult-content artifacts**.
- Applying digital forensic methodologies including registry hive analysis, VPN configuration discovery, and user attribution profiling, correlated dual user accounts (Eddie and Jimmy) to distinct roles in fraud, DRM circumvention, and anonymization, directly supporting federal charges under **CFAA**, **Wire Fraud**, and **DMCA** statutes.

Over Voltage Protection Using FET circuit

- Designed a circuit that safeguards electronic devices from potential damage caused by **excessive voltage levels**. The circuit effectively detects and diverts excess voltage away from sensitive components, offering reliable protection against **voltage spikes or surges**.
- Implemented an N-channel **MOSFET-based** switching mechanism that automatically disconnects the load when input voltage exceeds the predefined threshold, ensuring precise and fast response during transient conditions.

ACHIEVEMENTS

C2C CTF 2025 Finalist – International Capture The Flag (CTF) Competition

- Finalist in the global C2C CTF organized by **INCS-CoE** and hosted by **Northeastern University**, solving challenges in cryptography, OSINT, web exploitation, reverse engineering, and forensics while demonstrating expertise in adversarial tactics, red teaming, and offensive security assessments through collaborative multi-domain engagements.

CERTIFICATIONS

- CodePath CYB102: Intermediate Cybersecurity**
- Junior Penetration Tester (eJPT) - INE Security.**
- Certified Cloud Associate (ICCA) - INE Security**
- Offensive Hacking Unfolded - The Beginner’s Edition - comproavi.**
- Cyber Threat Intelligence 101 - arcX**